

1. Objective

Reconfigurable FPGA radar target detector

Official title: Design and Implementation of a Reconfigurable FPGA-Based Architecture for Real-Time Radar Signal Processing and Intelligent Target Detection.

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Team name: Beyond Boolean. **Platform:** PYNQ-Z2 / xc7z020clg400-1. **Tool:** Vivado 2025.1.1.
Processing clock: 50 MHz.

Implement a streaming cell-averaging constant-false-alarm-rate (CA-CFAR) style threshold detector in programmable logic. Demonstrate exact target detection, runtime sensitivity selection, valid-sample stalls and overflow-safe comparison using controlled synthetic magnitude sequences.

The FPGA generates and processes every sample; the host starts frames and reads retained results through Vivado VIO/JTAG. The scope is a reusable detection stage. RF acquisition, ADC, range/Doppler FFT, trained ML and partial bitstream reconfiguration are not implemented.

Verified outcome	Measured or checked result
Core simulation	10,721 checked results from 12,963 accepted samples
Controller and physical sweep	All 32 scenario/alpha/gap combinations plus repeat
Standard frame	80 full windows; targets at CUT indices 30, 60, 85
Routed timing	WNS 12.176 ns; WHS 0.018 ns; WPWS 2.000 ns

The original successful build and physical evidence are retained. This documentation update does not claim a new board execution. The original routed reports, checkpoint and native .xpr/IP source archive are now included. The final narrated video URL is intentionally blank.

2. Block diagram and detection method

PYNQ-Z2 radar target detector

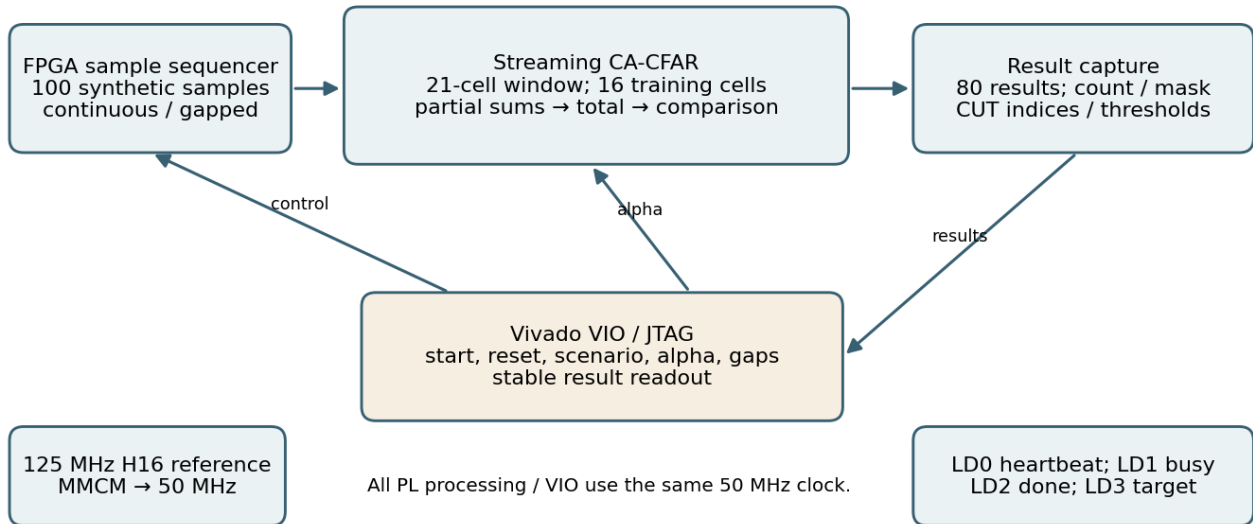


Figure 1. Source-derived architecture. The VIO, controller and detector share the 50 MHz clock. The H16 reference is nominally 125 MHz; the MMCM settings multiply by eight and divide by twenty.

A chronological window contains **8 training + 2 guard + 1 CUT + 2 guard + 8 training** samples. The sixteen training values are accumulated at full precision. Guard cells exclude nearby energy from the estimated background.

```

threshold = floor(training_sum / 16) * alpha
alpha = 1, 2, 4 or 8
detection = (unsigned CUT > full 19-bit threshold)
  
```

The total training sum has 20 bits; the full threshold has 19 bits. The displayed 16-bit threshold saturates at 65,535, while detection always uses the full threshold. Equality does not detect. These alpha values demonstrate sensitivity selection, not a calibrated probability of false alarm.

The window advances only on accepted samples. The first complete window uses samples 0-20 and reports CUT index 10. A 100-sample frame yields exactly 80 results, indices 10-89; no boundary padding is added. The controller drains the pending pipeline before finishing.

Partial training sums, total sum and threshold/comparison form three register stages. A result appears two clock intervals after accepting the newest sample in its full window. Configuration and CUT metadata travel with the arithmetic. Reset clears the window and pending valid results.

3. RTL design, hierarchy and schematic

Module / instance	Responsibility
radar_top	Board clock/MMCM, startup reset, VIO, LEDs and u_demo
radar_top.u_demo : radar_demo	Generate 100 samples; latch configuration; insert optional gaps; capture counts, masks, indices and timing
radar_demo.u_core : cfar_core	21-cell window, pipelined training sums, full-width threshold and comparison
radar_top.u_vio : vio_radar	Five control outputs and seventeen monitored inputs; generated by Vivado Tcl

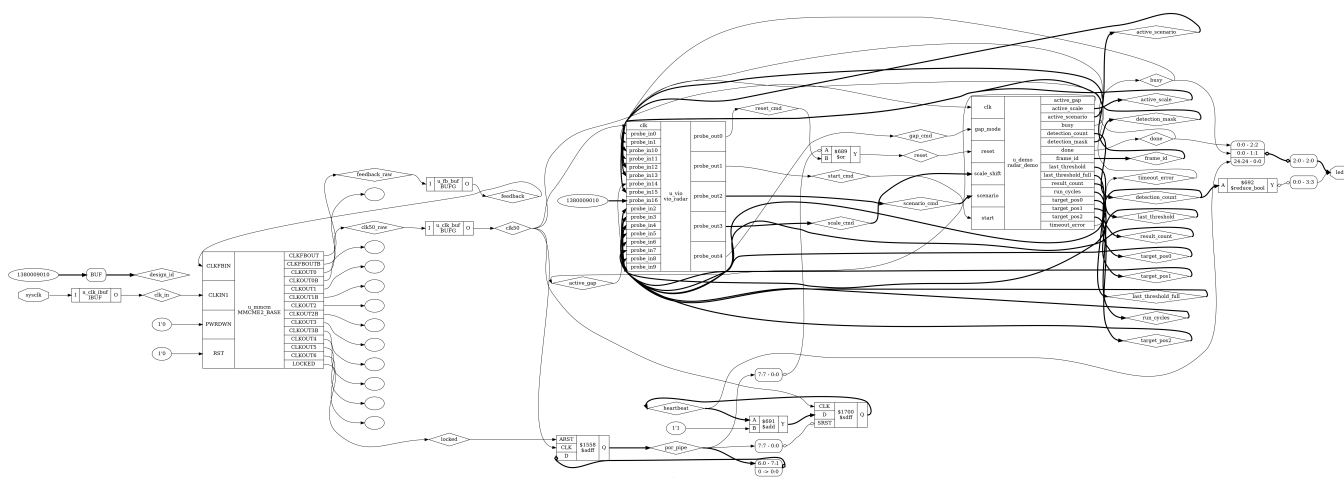


Figure 2. Genuine Yosys 0.33 elaborated top schematic. The full-resolution SVG/DOT, used-module JSON netlist and tool transcript are supplied for zoomed inspection. This overview is not a Vivado GUI export. Proprietary VIO and clock primitives are black boxes/library declarations in this offline structural check; their timing/behavior is established by the supplied Vivado build and physical evidence, not by Yosys.

Control and observation: reset, start, scenario, scale and gap are VIO outputs. Monitored values include busy/done, timeout, frame ID, counts, three positions, full detection mask, full/display thresholds, cycles, retained configuration and the RAD2 design identifier (0x52414432).

Clock/reset and constraints: 50 MHz clock buffers feed all processing and VIO. The eight-stage startup reset releases after clock lock. XDC assigns H16 and LEDs R14/P14/N16/M14, with LVCMOS33. Only human-visible LED output paths are false-pathed; detector/VIO internal paths are not exempted.

All three original RTL modules, both testbenches and all five original Vivado Tcl files retain exact byte/CRC identity with the successful build manifest. No functional RTL edit was made for this documentation upload.

4. Simulation results

The independent chronological-array scoreboard checks arithmetic, CUT indices, thresholds, valid timing and detections rather than reproducing the DUT shift-register implementation. Supplied Icarus Verilog 12.0 results passed on 8 October 2026. The full successful laptop log independently records both suites passing in Vivado XSim 2025.1.1.

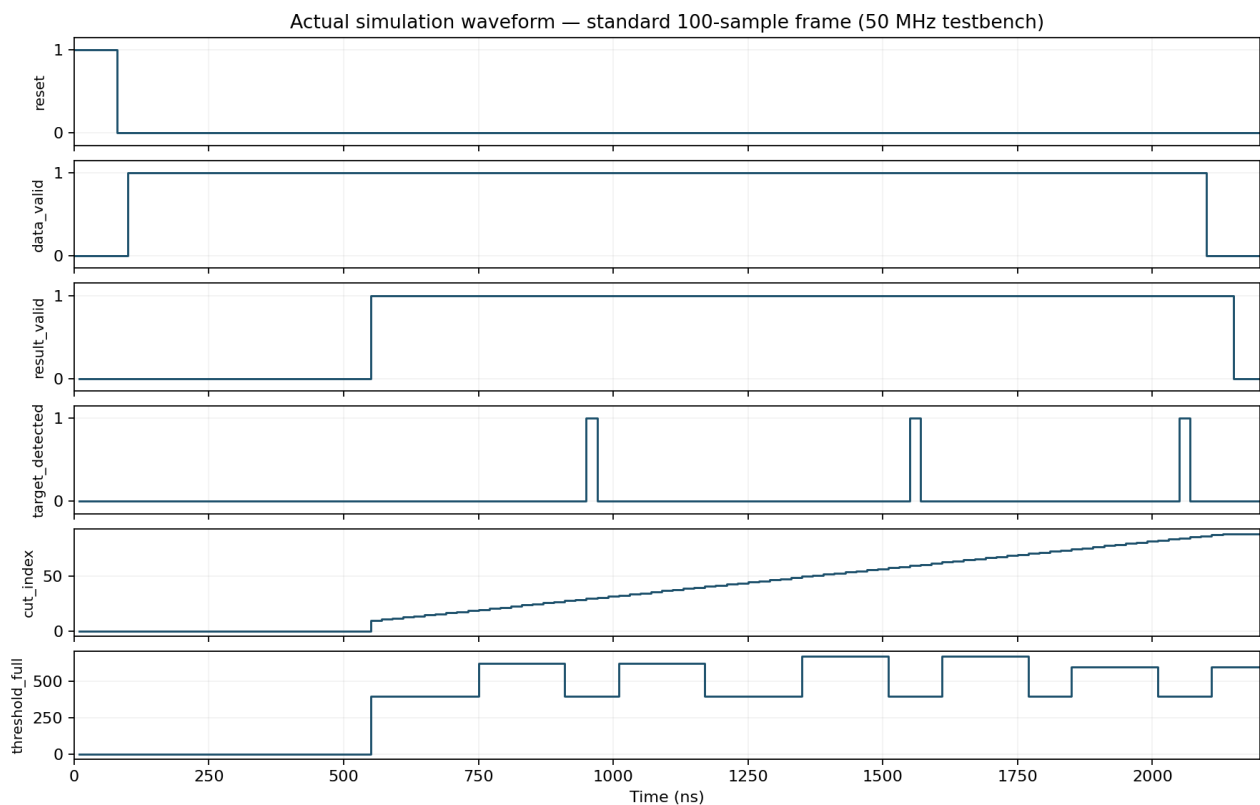


Figure 3. Actual supplied core VCD rendered as a waveform; not a fabricated GUI capture. The original VCD remains in Simulation/. See the dedicated simulation PDF for the standard-target plot, exact transcript and reproduction steps.

Test	Coverage / outcome
Core scoreboard	10,721 valid results; 12,963 samples; 112 directed/random/reset frame sequences
Core corner cases	Unsigned overflow, display saturation, equality, guards, truncation, gaps, random alpha changes, short frames and reset
Controller	4 scenarios x 4 alpha factors x 2 gap settings, then a repeat: 33 checked frames
Controller robustness	Held-start and reset during active frame checks
Standard synthetic sequence	Three detections at 30, 60, 85; 80 results for indices 10-89

Local simulation traces/logs are preserved under their real origin. The original Vivado log is Evidence/vivado_build.log. No additional simulator run is claimed during publication.

5. Hardware implementation - routed build

The supplied laptop build completed on **8 October 2026 at 22:40:16 IST**. It created a fresh xc7z020clg400-1 project, generated VIO, passed both XSim suites, synthesized, routed, checked setup/hold/pulse-width timing and DRC, and wrote matching bitstream/probe files. The full successful log and original BUILD_SUCCESS marker are included.

Metric	Actual available evidence
Vivado / device	2025.1.1 / xc7z020clg400-1
Processing clock	50 MHz; 20 ns period
WNS (setup slack)	12.176 ns
WHS (hold slack)	0.018 ns
WPWS (pulse-width slack)	2.000 ns
Bus skew	All four constraints MET; minimum slack 19.034 ns
Internal timing coverage	0 unconstrained internal endpoints
Routed DRC	0 Errors / Critical Warnings; 5 Warning checks
LUT / FF / BRAM / DSP	2010 LUTs / 3329 FF / 0 BRAM / 0 DSP; includes VIO/debug hub
Power	No measured power supplied; default-activity estimate not treated as measurement
Hardware outputs	Original radar_top.bit and radar_top.ltx, unchanged bytes

Recovered report review: all four bus-skew constraints pass. DRC reports three PDCN-1569 and one RTSTAT-10 warnings inside vendor dbg_hub, plus ZPS7-1 because this PL demonstration omits PS7. Methodology reports four LUTAR-1 async-reset warnings in debug-hub FIFOs. These warnings are disclosed, not claimed resolved. Retain the tested normal PYNQ boot setup. Check-timing reports zero unconstrained internal endpoints; only four intentionally false-pathed LED ports lack output-delay constraints.

Reports/ preserves the original eight report texts and routed checkpoint. FPGA_Project/ includes the successful native .xpr, VIO configuration/generated dependencies and original RTL/testbench/Tcl files, retaining their relative layout. Failed runs and disposable caches are excluded. Original Tcl remains the clean reproduction route; the tested BIT/LTX can be programmed without rebuilding.

An archive audit checks all eleven source CRCs and exact BIT/LTX SHA-256 identities, every saved board CSV row and the simulated standard-frame indices. Tools/validate_evidence.py records that audit; it does not rerun or remeasure hardware.

5. Hardware implementation - physical setup

The actual PYNQ-Z2 was programmed through Vivado Hardware Manager using the supplied matching BIT/LTX. The existing normal power/boot setup and USB/JTAG connection were used. Ethernet is visible in the photographs; the CFAR data/control path is in PL and VIO/JTAG, not network sample transport.



Figure 4. Original user-supplied PYNQ-Z2 setup photo from the Project 2 Drive folder. No generated hardware photograph is used.

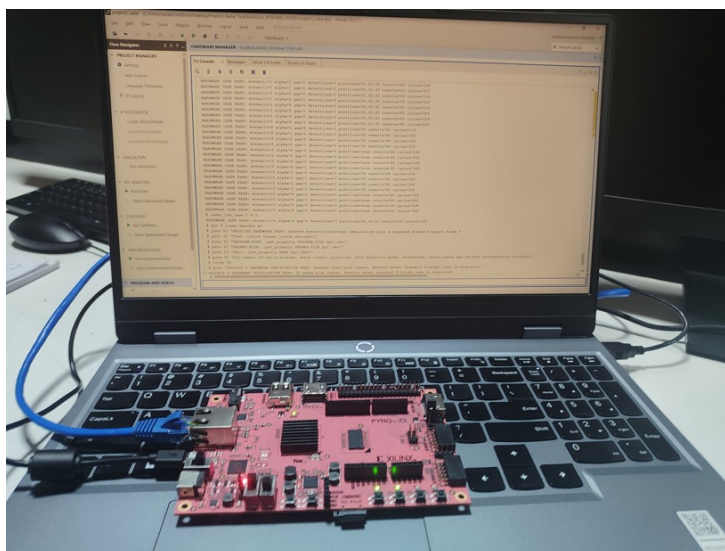


Figure 5. Original photo of the real board beside the laptop showing the passing Vivado hardware console. The exact console capture is supplied on the following page and in Images/.

5. Hardware implementation - verified output

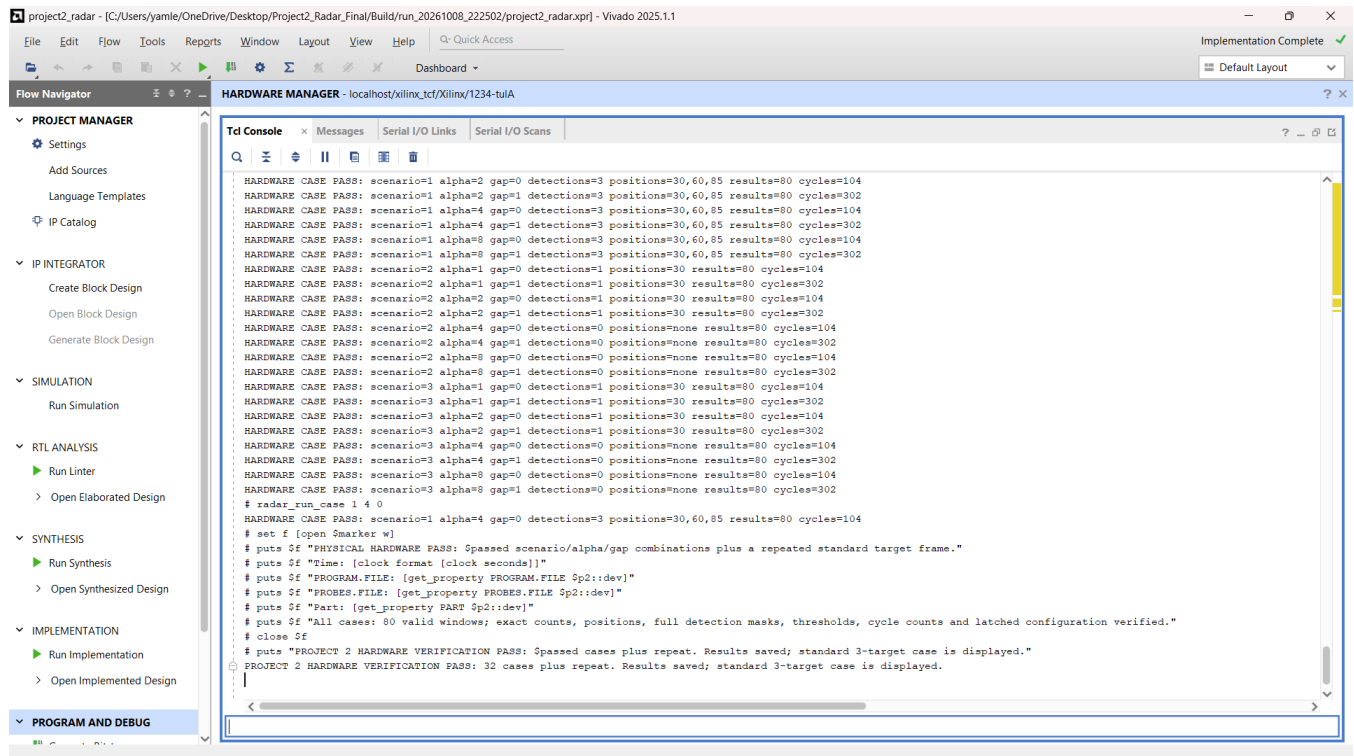


Figure 6. Actual Vivado console: all combinations passed and final standard three-target frame retained. Original HARDWARE_PASS marker time: 8 October 2026 23:54:05 IST.

Physical case	Alpha	Targets / zero-based indices
Noise: constant 100	1 / 2 / 4 / 8	0 / none
Targets 1000, 1200, 900; background 100	1 / 2 / 4 / 8	3 / 30, 60, 85
Weak 300; background 100	1 / 2	1 / 30
Weak 300; background 100	4 / 8	0 / none
Target 60000; background 20000	1 / 2	1 / 30
Target 60000; background 20000	4 / 8	0 / none; full-width threshold

Every scenario/factor pair passed with both continuous and gapped input: **32 unique combinations**, followed by a repeated standard frame. The unchanged saved CSV contains **39 PASS rows**, including additional repetitions/demonstrations. Reset explains repeated frame IDs in the later demonstration rows.

Each frame returned 80 windows. Continuous input takes 104 cycles (2.08 microseconds); two idle clocks between samples takes 302 cycles (6.04 microseconds). These are FPGA frame-processing times at 50 MHz, excluding host/JTAG command overhead.

The hardware script verifies counts, three positions, the full 128-bit detection mask, thresholds, cycles, frame/configuration and design ID before writing a PASS row. The CSV itself stores counts/cycles/last threshold, not separate position/mask columns; the PASS marker and console corroborate those additional checks.

6. Applications

A streaming background-adaptive threshold stage can follow a radar magnitude or power pipeline and mark candidate detections. Deterministic FPGA processing and one accepted sample per clock support integration with streaming acquisition/FFT logic. This controlled experiment establishes the detector arithmetic, valid timing and observation interface rather than a full sensing system.

Runtime alpha selection demonstrates the sensitivity tradeoff: the weak target at index 30 is rejected at factor four and accepted at factor two. The same hardware and bitstream handle both choices. Gapped input preserves the exact detections because the sample window advances only on `data_valid`.

7. Conclusion and future scope

The original complete design passed the independent core simulation, all controller cases and the physical 32-combination sweep plus repeat. The standard frame produced three correct indices and 80 full windows. Routed setup/hold/pulse-width margins are positive at the 50 MHz processing clock; the original bitstream and matching probes are preserved.

The original utilization, timing, methodology and bus-skew reports have been reviewed. Future work includes reviewing debug-IP warnings for deployment; integrating real acquisition and range/Doppler processing; calibrate alpha against representative clutter and false-alarm requirements; and validate performance on recorded or live radar measurements. A wider framing/index scheme would be needed for long streams beyond the demonstration's 100-sample frames.

Submission status: final narrated video URL intentionally blank. Team: Beyond Boolean. All required Project 2 technical-document categories are supplied, including original routed reports and the native project source archive. Completion of the final video and its accessible Drive link remains pending.

Evidence and primary references

Local evidence: Results/BUILD_SUCCESS.txt; Results/HARDWARE_PASS.txt; Results/hardware_results.csv; Evidence/vivado_build.log; Reports/*.rpt and routed_design.dcp; FPGA_Project/Project2_Radar_Vivado_Source.zip; Simulation/transcript.txt and VCD/CSV files. The readiness audit maps every organizer item to a file.

AMD/Xilinx PYNQ-Z2 XDC: github.com/Xilinx/xup_fpga_vivado_flow, source/pynq-z2/lab5/uart_led_pins_pynq.xdc. AMD 7-Series Libraries Guide UG953: MMCME2_BASE. AMD Vivado Tcl Command Reference UG835: `commit_hw_vio` and `refresh_hw_vio`. Original links are in README.md.